

JORDAN NEWMAN

Senior Software Engineer | Systems, Backend, Security, Infrastructure

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SUMMARY

Senior software engineer specializing in systems software, networking, security, distributed infrastructure, and performance engineering. Built production software spanning web servers, packet analyzers, authentication systems, kernel modules, deployment automation, and enterprise integrations. Experienced in C, C++, Go, Python, C#, FreeBSD, Linux, Azure DevOps, Docker, Kubernetes, Terraform, and CI/CD.

TECHNICAL SKILLS

Languages: C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic

Systems: FreeBSD, Linux, kernel modules, system calls, bootloaders, operating systems, kqueue, GDB, TCP/IP

Networking/Security: DDoS detection/mitigation, packet capture, DNS analysis, libpcap, IPFW, authentication, telemetry

Embedded: Embedded C, Atmel AVR, SPI, UART, I2C, Arduino, circuit design, soldering, hardware troubleshooting

Cloud/DevOps: Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation

Databases/Web: MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Built and operated infrastructure spanning 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps peak traffic.
- Developed production C web server supporting 150,000+ concurrent connections and high-traffic customer workloads.
- Built DDoS mitigation systems, packet analyzers, authentication systems, telemetry, and security platforms at ISP scale.
- Implemented hands-on Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows.
- Developed production FreeBSD kernel modules, embedded firmware, and low-level systems software.

SELECTED SYSTEMS PROJECTS

TAFOS Operating System: Developed educational x86 OS kernel in C and assembly with custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

KeepClean: Developed FreeBSD kernel module hooking syscalls to monitor execution activity, protect system assets, and report telemetry to userspace services.

VirtualDir: Developed FreeBSD kernel module remapping filesystem paths through syscall interception; deployed customer-facing across approximately 30 production servers over 5-8 years.

StatCache: Developed FreeBSD kernel module instrumenting stat() and lstat() activity for filesystem visibility and performance analysis.

PrimeDump: Developed libpcap-based traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 headers with ncurses real-time views and IPFW integration.

PrimeDNSTop: Developed DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source/domain activity.

AVR Smart Smoke/CO2 Detector: Developed contract firmware in embedded C for battery-powered wireless Atmel AVR prototypes using SPI, UART, and I2C interfaces.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist | Remote / Tampa Bay, FL | July 2022 - Present

- Implemented Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.
- Partnered with architecture teams to standardize deployment practices, DevOps policies, and production software delivery.
- Developed internal AI workflows, reusable libraries, and documentation to improve engineering productivity and consistency.

DDI Systems - Senior Development Specialist | Manalapan, NJ | November 2021 - July 2022

- Modernized legacy VB.NET software by porting components to C#, improving maintainability and long-term reliability.
- Integrated Microsoft Office 365 SMTP OAuth, replacing deprecated authentication with secure customer-facing workflows.

ISPrime LLC - CEO | Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization, operational cleanup, and software platform improvements for production hosting infrastructure.
- Refined internally developed monitoring, security, and infrastructure tools to improve reliability and operational efficiency.

DDI Systems - Computer Programmer | Manalapan, NJ | April 2017 - April 2018

- Integrated ERP software with SellerCloud and Essendant systems, automating order, catalog, image, and SDS data workflows.
- Developed customer-facing import tools for MultiValue/Universe databases and SFTP-based supplier integrations.

Too Much Media LLC - Computer Programmer | Morganville, NJ | January 2016 - April 2017

- Developed real-time browser chat integrated with ticketing systems, improving customer support responsiveness.
- Implemented OAuth 2.0 authorization for protected websites and customer-facing access workflows.

MFCXY, Inc. - Computer Programmer | Chicago, IL | December 2014 - September 2015

- Developed backend server features and Windows client integrations while identifying performance and security issues.
- Built middleware to migrate data across database platforms and improve operational continuity.

AJPM, LLC - Programmer / Owner | *New Jersey / Remote* | *November 2012 - December 2013*

- Built automated purchasing and inventory software tracking acquisition, repair, resale, and financial workflows.

ISPrime Inc. - CIO / Partner | *Weehawken, NJ* | *March 2001 - December 2014*

- Built and operated infrastructure spanning 3,000-4,000 servers across 10 CDN locations and multiple datacenters.
- Sustained peak traffic exceeding 65 Gbps while supporting mission-critical customer workloads.
- Developed PrimeHTTPD, a high-performance C web server optimized around FreeBSD kqueue supporting 150,000+ concurrent connections.
- Deployed PrimeHTTPD across approximately 200 production servers supporting high-traffic customer environments.
- Developed DDoS detection and mitigation systems protecting infrastructure during attacks involving up to 750,000 compromised hosts.
- Created PrimeDump and PrimeDNSTop, libpcap-based network analysis tools for traffic anomalies and DNS attacks.
- Developed production FreeBSD kernel modules including KeepClean, VirtualDir, and StatCache for security and filesystem observability.
- Built monitoring, telemetry, authentication, and security systems used across thousands of production servers.

FatWallet.com - Systems Administrator | *Wisconsin / Remote* | *December 2000 - June 2001*

- Rebuilt production infrastructure to resolve performance and stability issues without customer-facing downtime.
- Built automated alerting systems for website availability, server health, and operational response.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.